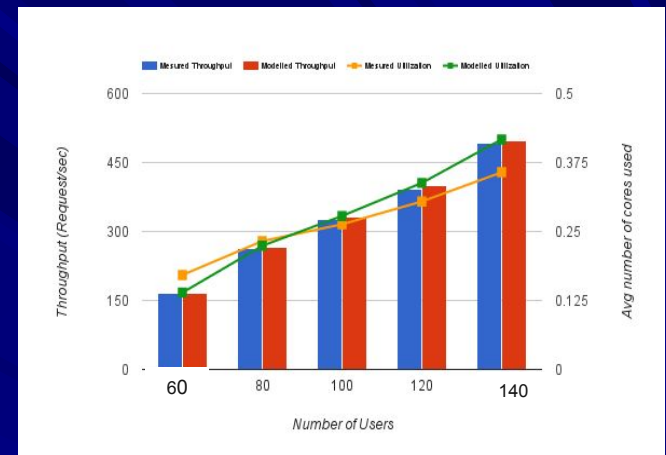
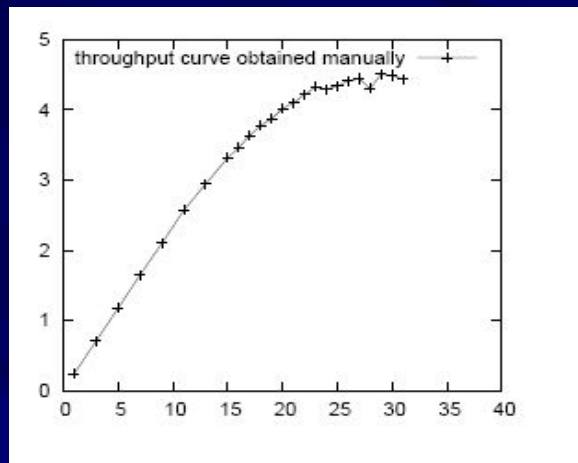
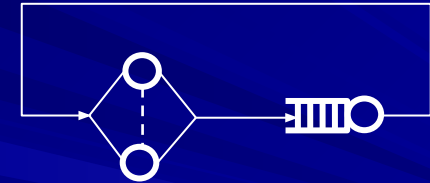
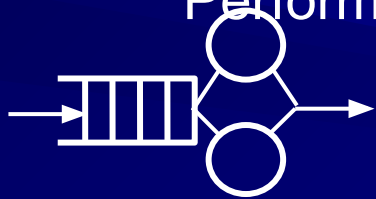




CS 681-2025

Performance Analysis of Computer Systems and Networks



Prof. Varsha Apte

Department of Computer Science and Engineering,
IIT Bombay



Course Introduction



Computer Systems/Network Performance

"Performance" of Computing Systems and Networks relates to qualities such as:

- the total time required to finish some task, as perceived by the user of the system

e.g. *file downloading delay*,

- the *amount* of work per unit time that the system itself is doing
e.g. Number of trades completed per hour at peak hour by a stock exchange system

Good performance is a critical quality requirement of all computing and network systems

Nonetheless, we often run into "slow" performance



Figure Credit[11]

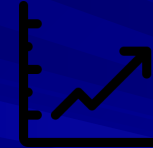


Figure Credit[12]

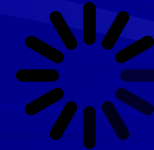


Figure Credit[10]

The Dreaded Loading Icon...



CS 681-2025
Computer and Network Performance Analysis

Who What Why How ...

About the course



What: is the course about?

In this course we focus on *analyzing* performance of systems.
The fundamental question we answer is:

Given some essential description of a computing or networking system - can we estimate its "performance metrics"?

e.g. can we estimate download delay of a file storage application?

Can we estimate whether a stock exchange server can carry out some X number of trades per hour?



Why: are these questions important?

- For businesses: to know whether their users will be satisfied with their service
- If not, how can they improve performance?
 - Can help determining 'sizing' of resources. E.g.
 - What should be the bandwidth of the link coming into a data center?
 - How many cores should a server have?
 - What exactly is the 'bottleneck'?
 - Can help select scheduling policies - e.g. Round robin or FCFS?



How: Queuing Theory

View all resources as "queuing systems"

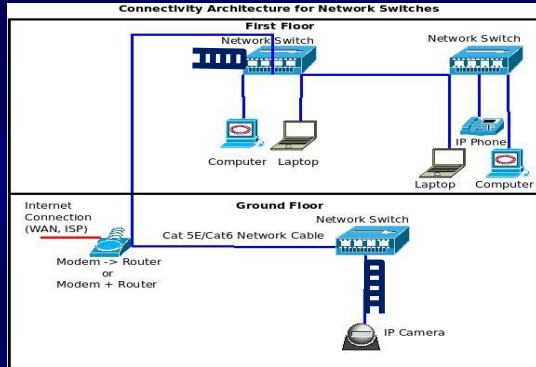
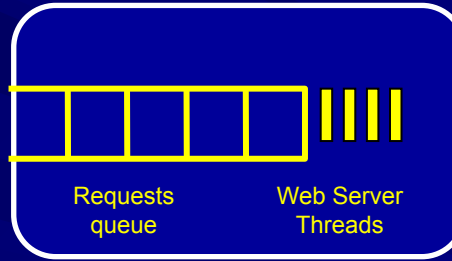
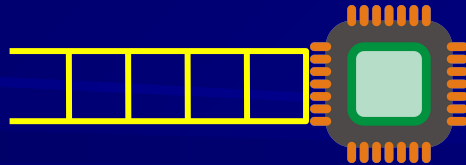


Figure Credit[3]

Packet queue for network links



Web requests queue for Web server



Job queue (processes or threads) for processor

Figure Credit[5]

A universal mathematical theory that models shared resources as 'queuing systems'

- 'Laws' that relate performance metrics with system parameters with wide applicability
- 'Formulae' that express a performance metric in terms of system parameters

How: Discrete Event Simulation

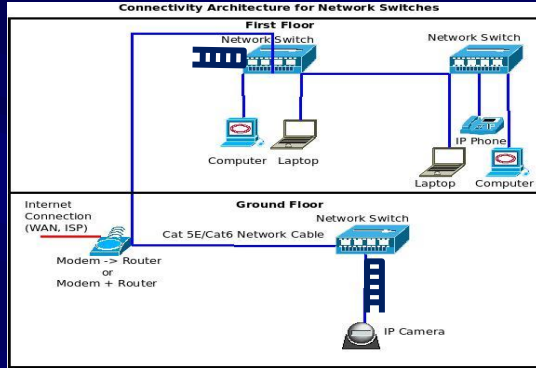
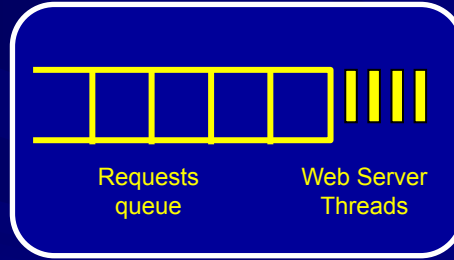
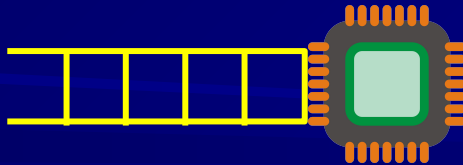


Figure Credit[3]

Packet queue for network links



Web requests queue for Web server
Web Server Class: 'queue' data structure, 'thread pool' member variable... etc



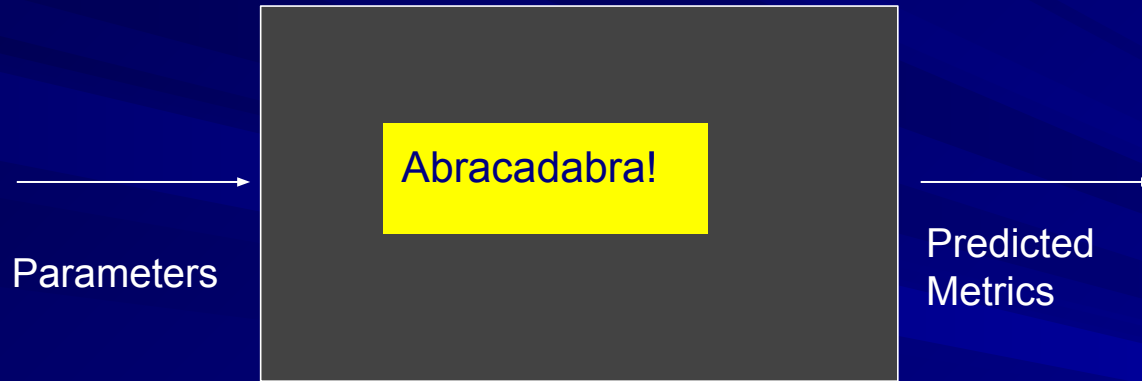
Job queue (processes or threads) for processor

Figure Credit[5]

A computer program that represents entities of the actual system with objects or data structures in a program

- E.g. a processor object, a queue object etc
- Events (e.g. job arrivals) are generated and processed in the program and performance metrics collected

How: Machine Learning Models



- Increasingly, ML models are being used to predict system performance, especially for complex systems with large number of parameters



What: Is the Syllabus?

Performance measures of computing and networking systems; open and closed loop load testing and performance measurement; performance modeling using Markovian queuing systems (M/M/c/k), asymptotic analysis of queues, basic queuing system laws: Utilization Law, Little's Law; non-Markovian queues (M/G/1), open and closed queueing networks; design, implementation and statistical analysis of discrete event simulation models; application of data driven models for performance prediction: (machine learning models- e.g. support vector regression, clustering based models, Bayesian optimization, neural network models). Case studies and applications of these methods through paper-readings and projects in a spectrum of distributed and networked systems --- e.g., web systems, virtualization platforms, clouds, distributed data processing platforms and learning systems, distributed applications, networking protocols, network backbone architectures, etc



What is NOT the syllabus

- Details of various mechanisms of systems or networks
- Study of how to **design** protocols, scheduling disciplines, any specific functionalities etc for improving performance
 - (Focus is **ANALYSIS**, not design)
- Advanced mathematical models in queuing theory



What: are the pre-requisites

- Undergraduate background (high level) in:
 - Operating Systems
 - Computer Architecture
 - Computer Networks
- 12th standard level Mathematics
- Probability and distributions (some review will be done)
- Proficiency with using and programming machine learning models: this will not be taught.



Who: Should take this course

- Those interested in 'systems': operating systems, Networked applications, virtualized systems, cloud computing
- Those interested in Networks: Wired networks, Wireless networks, etc.

Queuing Systems is foundational theory for understanding these systems



How: will you be assessed

- Two quizzes + Midsem 'quiz' (Best two): $2 \times 11 = 22$
- Three hands-on assignments (possible major evaluation in lab/classroom quizzes) : $3 \times 14 = 42$. *To be done in Teams of two.*
- (Theory) Endsem: 26
- Attendance and Participation: 10

*Assessment likely to be severely affected by class size and lack of available TAs. **Old style (I did all assignment vivas) will not work.***

Quizzes may have many objective, auto-graded questions

Assignments will be evaluated in same scale-friendly way (objective, auto-graded questions based on labs done)

Please drop the course if you do not like this.



What is the attendance policy?

- No DX
- Attendance + participation: 10% weightage
- Low attendance —> forgo direct interaction with me, maintain consistency as a 'remote' student
- 'Participation' - in-class, online
 - Homeworks will be posted every week, (if class stays large) will be 'assigned' to a group of students for half the week, then 'free for all'



What: is Participation 10% based on

- Attendance (even though no DX, attendance will be recorded, and will contribute to participation marks)
- In-class participation (answering questions, asking questions)
- Online participation
 - Homework will be released to a specific group every week
 - That group gets 3 days to answer - the group must co-ordinate and assign questions to be answered
 - After 3 days anybody can answer (*tentative policy, may change).



What: are the Hands-on assignments

1. Assignment 1: Performance measurement of a simple script on a Web server ('load testing' and analysis of results)
2. Assignment 2: Discrete event simulation modeling of a Webserver with some complex features modeled
3. Assignment 3: Machine learning model of a Web server running in a more complex environment (virtualized, etc)

All assignments are self-driven. Some doubt clearance to the extent possible by TAs, but these are not like UG labs, there will be no hand-holding.



What: is the GenAI, copying and other academic ethics policy

- GenAI in homeworks, assignments - not recommended, but no way for us to check.
 - Any usage will be required to be clearly specified
 - Part of the assignment evaluation will be in class/lab with 'exam like' controls.
 - In vivas, if lack of understanding of code is displayed
 - If after honest declaration of AI usage, marks will be cut
 - If without declaration of AI usage - reported to DADAC
- In general, high academic ethics is expected as per usual rules, violations reported to DADAC



How/What/Who: misc :)

- Course Management Platform (for uploads etc): Bodhitree
- Course Material:
<https://www.cse.iitb.ac.in/~varsha/cs681.php>
 - *Detailed Lecture Notes*
 - *Class is/was BOARD writing based (never been based on slides)*
 - *However, slides and videos available for NPTEL version*
- Office hours: Will announce after schedule settles. Till then, ping on MS Teams and come to KR 308
- Currently two PGTAs: Chaitanya Shinge, Shan Muhammed



Thank You.



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